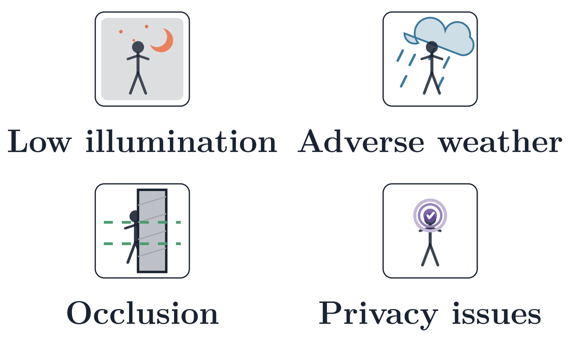
Few-Shot Radar Person Identification via Tracking-Integrated InISAR and Pre-Trained Meta-Learning

Jeffee Hsiung* Rengin Torun* S. Hamed Javadi† Hichem Sahli†
KU Leuven | imec | Vrije Universiteit Brussel *Equal contribution (first authors) †Equal contribution (supervisors)

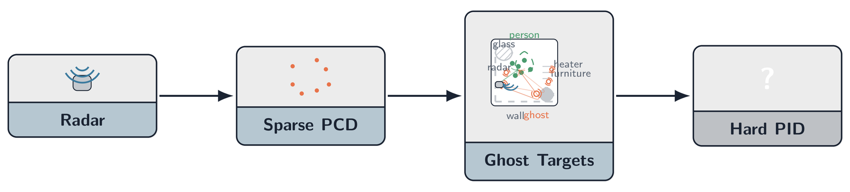
❶ Introduction

Vision-based person identification (PID) struggle under conditions [1, 2]:



Radar Advantages: Zero optical exposure · All-weather operation · Penetrates thin barriers · Privacy preserving

❷ Challenges in Radar-Based PID



Radar point clouds are sparse, noisy, and contaminated by multipath ghosts

Low Angular Resolution
Conventional angle-of-arrival [3] yields motion-defocused clouds and multipath artifacts.

Rigid-Target Bias
Interferometric Inverse Synthetic Aperture Radar (InISAR) improves resolution but is validated only on large *rigid* targets [4, 5].

Multipath Ghosts
Ghosts overlap in range-Doppler and mimic real 3D clusters [6]

Data Scarcity
Existing SOTA [7, 8] demand > 100 seq/person; Point Cloud Transformer (PCT) [9] handles spatial structure but targets *static* objects [10].

Gap: Tracking-integrated InISAR + pre-trained meta-learning.

❸ Key Results

73.8% InISAR IoU

< 5.5 cm Height Error

70.0% Cross-Session PID

Contributions & System Impact

Articulated Validation: First InISAR validated on humans in multipath-rich indoor envs.

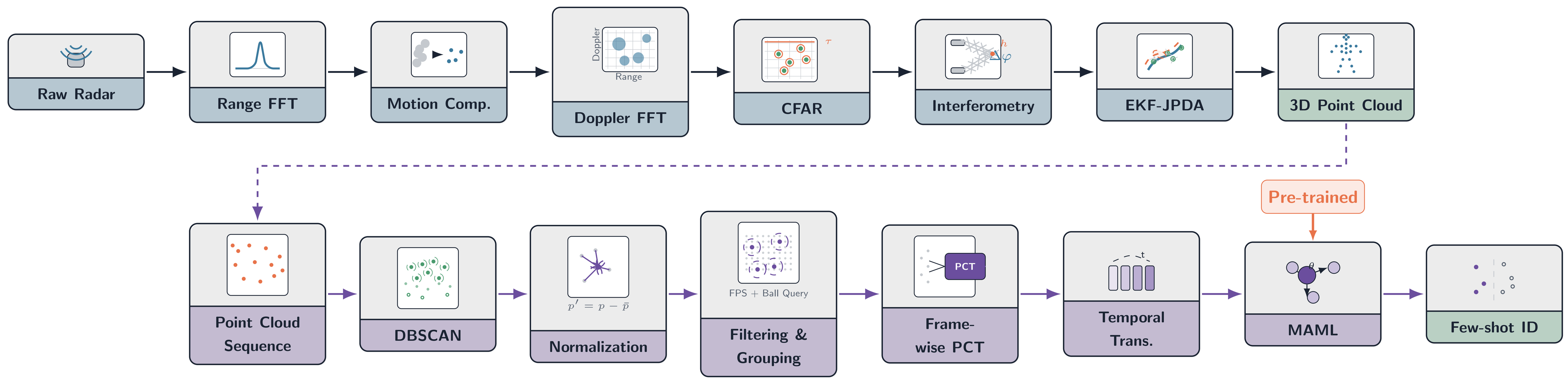
Resolution-level Error: < 5.5 cm (med. 3.87 cm) approaches Δr=4.55 cm limit (vs. prior 0.2–1.6 m errors).

SOTA Supervised: PCT-Trans achieves 98.99% supervised (+2.81pp over TCPCN).

MAML Gains: Pre-trained MAML yields 70.0% cross-session (+15.6pp over random).

Deployable few-shot radar PID with minimal training data.

❹ Our Framework: Unified Sensing + Meta-Learning



End-to-end system: InISAR 3D reconstruction → spatio-temporal learning → PID

(1) Radar domain
Tracking-integrated InISAR reconstructs high-res 3D point clouds of articulated human motion, validated against motion-capture ground truth.

(2) ML domain
Spatio-temporal transformer (PCTtrans) embedded in Model-Agnostic Meta-Learning (MAML); task-specific pre-trained initialization substantially improves few-shot adaptation.

(3) System integration
High-quality 3D reconstruction + meta-learning together enable practical few-shot radar PID, reducing dependence on large subject-specific datasets.

❺ InISAR Pipeline: 3D Reconstruction

1. Range FFT

- FMCW chirp compression to $S_{R,m}(f,t)$ per m-th antenna.
- Physical range limit $\Delta r = c/(2B)$ *bounds our achieved height error.*

2. Motion Compensation (ICBA)

- Translational model $R_0(t) = r_0 + \beta t + \gamma t^2$.
- Nelder-Mead optimizes contrast $[\hat{\beta}, \hat{\gamma}]$ to remove smearing → yields *signal-to-noise ratio (SNR) enhanced* $S_{C,m}(f,t)$:

$$[\hat{\beta}, \hat{\gamma}] = \arg\max \left\{ \frac{\sqrt{\frac{1}{N}\Sigma(I_i - \bar{I})^2}}{\bar{I}} \right\}$$

3. Coherent Combination & Doppler FFT

- Geometric phase corrections align all MIMO virtual antennas (SNR boost) → *focused* 2D Range-Doppler Profile (RDP).

4. CFAR Segmentation

- SOCA-CFAR ($N_T = 6,\ P_{FA} = 10^{-10}$) *isolates target cells* from noise/clutter.

5. Interferometry

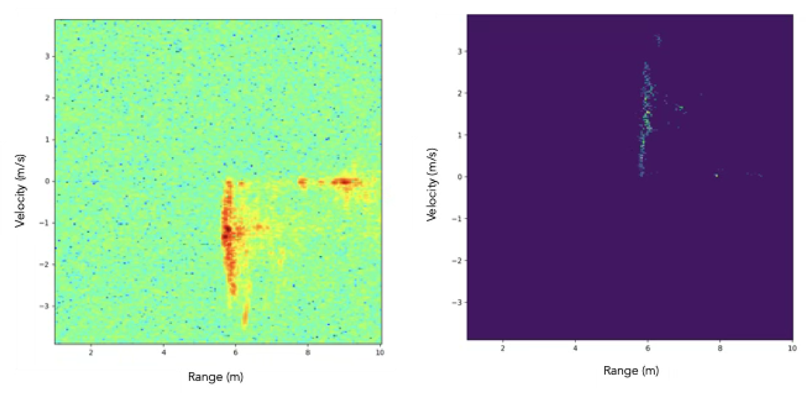
- Phase difference between top/bottom MIMO rows *yields elevation*:

$$I_1 \cdot I_2^* = |I_1||I_2| \exp\left(j\frac{2\pi}{\lambda}\Delta R\right) \quad \rightarrow \quad h = \frac{\lambda \phi R}{2\pi L}$$

- Exploiting MIMO antenna pairs reduces measurement noise.

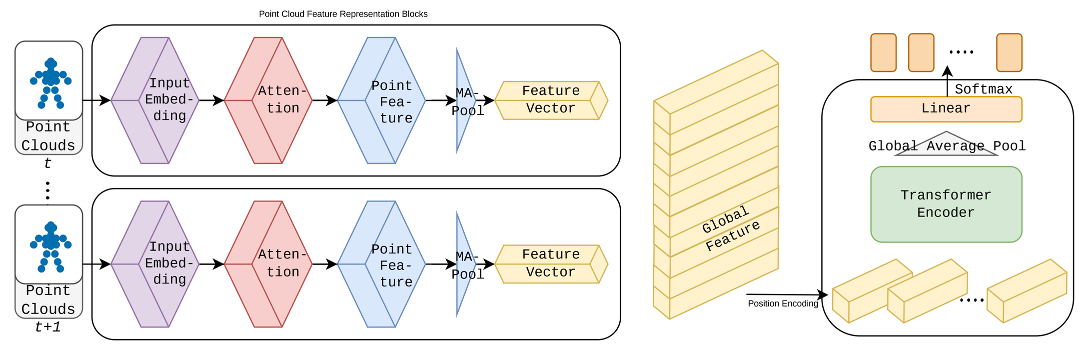
6. EKF-JPDA Tracking

- Spatial-domain JPDA handles data association; EKF filters state covariance.
- Max miss count=1 → *prunes multipath ghosts* while tolerating occlusions.



Raw RDP *Post-InISAR RDP*

❻ Few-Shot PID with PCTtrans



PCTtrans: Shared Spatial Extractor + Temporal Gait Model

Gait-Invariant Processing

- Centroid norm. $p'_i = p_i - (\bar{x}, \bar{y}, 0)$ prevents path overfitting

Spatial PCT Encoder

- Farthest Point Sampling → non-uniform density
- Ball Query → rotation-invariant neighborhoods.
- Neighbor Embedding: $\Delta F(p) = \text{concat}_{q \in \text{knn}(p,k)}(F(q) - F(p))$
- Offset-Attention acts as a discrete Laplacian: $F_{\text{out}} = \text{LBR}(F_{\text{in}} - F_{\text{sa}}) + F_{\text{in}}$ ($\mathbf{f}_t \in \mathbb{R}^{192}$)

Temporal Transformer

- Positional encoding injects order
- Multi-head self-attention captures cross-frame gait dynamics: $p(y|\mathcal{S}) = \text{Softmax}(W_2 \cdot \text{ReLU}(W_1\mathbf{g} + b_1) + b_2)$

❼ Pre-trained Meta-Learning [11]

Two-Phase Domain-Aware Adaptation

- **Phase 1:** Supervised pre-training (3 subjects) → θ_0^{PT} (session-invariant patterns).
- **Phase 2:** MAML meta-training [12, 13].

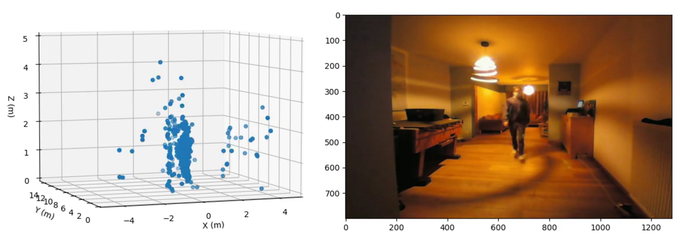
Inner: $\theta'_i = \theta - \alpha\nabla\mathcal{L}$ Outer: $\theta \leftarrow \theta - \beta\nabla\Sigma_i\mathcal{L}$

Key Insight: θ_0^{PT} lies closer to optimal θ^*, enabling faster convergence than random initialization.

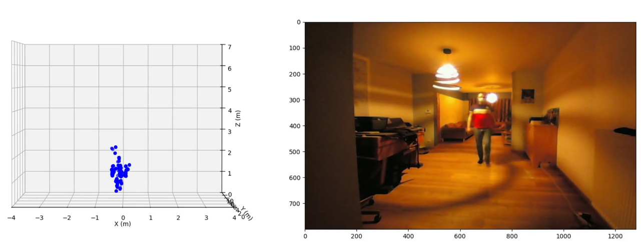
Gain: +15.6pp over random MAML

❽ Experimental Results

Ghost Removal (EKF-JPDA)



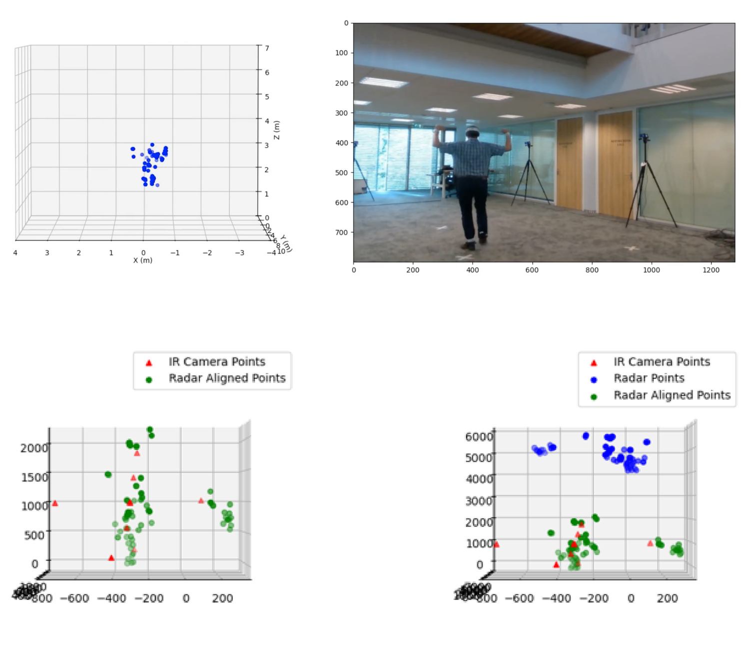
(a) Before: Ghost remains



(b) After: Ghost rejected

Scan to watch video demo

InISAR vs Motion Capture



Scenario	IoU (%)
Circular Walking	79.82
Hands-Up Walking	100.0
Kicking	69.19
Squatting	59.98
Average	**73.78**

Dataset: 5 subj. (164–185 cm), $B = 3.3$ GHz ($\Delta r = 4.55$ cm).

Reconstruction Height Error < 5.5 cm (median 3.87 cm)

PID Benchmark Accuracy (%)

Query Type	TCPCN [8]	PCT-T (sup.)	MAML (rand.)	MAML (pre-train)
Q1(same session)	96.18	**98.99**	76.67	94.44
Q2(cross-session)	35.10	32.05	54.44	**70.00**
Q3(cross-subject)	57.81	40.09	46.67	48.89

- **Supervised:** PCT-T outperforms TCPCN → **+2.81pp**.
- **Few-Shot:** Pre-trained MAML vs Random → **+15.56pp**.
- **mmGait Benchmark:** PCT-T leads by **+11.5pp** on fixed trajectories [14].

jeffee.hsiung@icloud.com · torun10@imec.be · hamed.javadi@imec.be · hichem.sahli@imec.be · EUSIPCO 2026, Bruges · imec / KU Leuven / VUB

Full references: scan QR

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